

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find the solution of the following initial value problem:

$$y'' + 8y' + 25y = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

Characteristic equation is

$$r^2 + 8r + 25 = 0$$

$$\Rightarrow r = -4 \pm 3i$$

Thus

$$y(x) = e^{-4x}(C_1 \cos 3x + C_2 \sin 3x)$$

So we get finally

$$\underline{y(x) = e^{-4x}(\cos 3x + \sin 3x)}$$

$$y'(x) = -4e^{-4x}(C_1 \cos 3x + C_2 \sin 3x) + e^{-4x}(-3C_1 \sin 3x + 3C_2 \cos 3x)$$

$$y(0) = 1 \Rightarrow C_1 = 1$$

$$y'(0) = -1 \Rightarrow -4C_1 + 3C_2 = -1 \Rightarrow \begin{cases} C_1 = 1 \\ C_2 = 1 \end{cases}$$

Problem 2. (5 points) Find the general solution of the homogeneous differential equations

$$9y^{(3)} + 12y'' + 4y' = 0.$$

Characteristic equation is

$$9r^3 + 12r + 4r = 0$$

$$r(9r^2 + 12r + 4) = 0$$

$$r(3r + 2)^2 = 0$$

$$\Rightarrow r_1 = 0, \quad r_2 = r_3 = -\frac{2}{3}$$

Thus

$$y(x) = C_1 e^{0 \cdot x} + e^{-\frac{2}{3}x}(C_2 + C_3 x) = C_1 + e^{-\frac{2}{3}x}(C_2 + C_3 x)$$